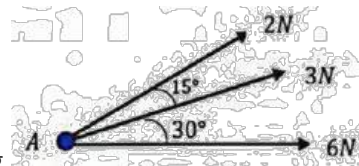


Motion and Forces Topic Test 1

Total Marks: 30

Time: 50 minutes

- 1 Forces of 6 N , 3 N and 2 N act on an object, considered point A , as shown in the diagram. The vector sum of these forces acting on the object at point A is called the resultant force. The force 6 N is acting along a horizontal plane.

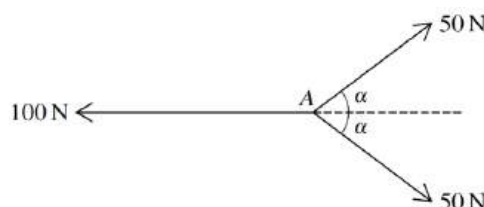


- (i). Find the magnitude of the resultant force, correct to one decimal place. 3
- (ii). Find the direction of this resultant force, correct to one decimal place. 1

- 2 Three coplanar forces of magnitude 100 N , 50 N and 50 N act at a point A , as shown in the diagram below.

The angle α is such that $\cos\alpha = \frac{4}{5}$. Find the magnitude

of the resultant of the three forces and state its direction.



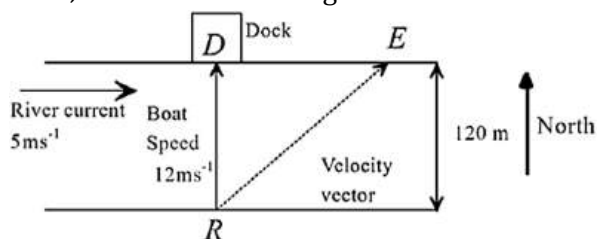
- 3 Rylie has a boat which moves at a top speed of 12 m/s in still water. From point R , he wants to go due north to point D on the opposite side of the river, as shown in the diagram below.

Today the current in the river is flowing at 5 m/s .

From R , he steers the boat due north toward D

at top speed. Due to the current, he drifts down

the river and arrives at point E .

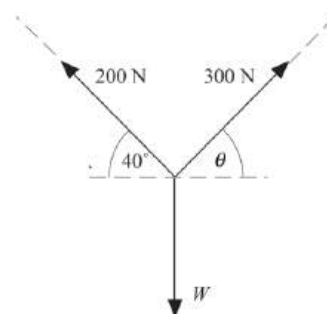


- (i). Taking R as the origin, write down Rylie's velocity vector in the form $x\mathbf{i} + y\mathbf{j}$ and find the magnitude of this vector. 2
- (ii). What is the bearing of Rylie's velocity vector and how far does he travel from R to E ? 2
- (iii). On what bearing should Rylie have pointed the boat, so that he arrived at D , with the boat travelling at its top speed? 2

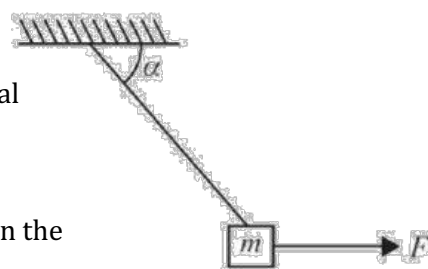
- 4 Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented by vectors $10\hat{i} + 20\hat{j}$, $20\hat{i} - 30\hat{j}$ and $-6\hat{i} + 20\hat{j}$ Newtons respectively.

- (i). Draw a sketch showing the relative magnitude and direction of the three forces. 2
- (ii). Determine the resultant force, expressed as a vector, and find its magnitude 2
- (iii). An additional force $\vec{F}_4 = k\hat{j}$ is now applied so that the resultant of $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ and $\vec{F}_4$ has magnitude $30N$. What is the value of k ? 2

- 5 The diagram shows the force of gravity, W , that pulls straight down on an object. Two forces of 200 N and 300 N are exerted by cables as shown. If the resultant force is zero, find θ and W .



- 6 A particle of mass m kilograms hangs from a string that is attached to a fixed point. The particle is acted on by a horizontal force of magnitude F newtons. The system is in equilibrium when the string makes an angle α to the horizontal, as shown in the diagram below. The tension in the string has magnitude T newtons. Find the value of $\tan\alpha$.



- 7 Two forces are acting on an object: 200 N acting at $N65^\circ W$ and 250 N acting at $N30^\circ E$. Calculate the resultant force $\vec{F}$ acting on the object, correct to one decimal place. 3

- 8 The position of jet ski relative to a stationary ship is plotted on the ship's radar. One unit on the radar represents 1 metre along the sea level, and the origin represents the ship's position. The jet ski travels at a constant speed and is observed at point A on a position vector of $-20\hat{i} + 36\hat{j}$. One second later, the jet ski is observed at point B on a position vector of $-2\hat{i} + 51\hat{j}$.

A crew member on the ship observes the motion of the jet ski using binoculars for ten seconds starting from point A , without taking his eyes off of the jet ski. 2

- (i). Find the position vector of the jet ski after the ten seconds. 2
- (ii). Hence find the angle through which the crew member's binoculars moves through during this ten second interval, correct to the nearest minute.

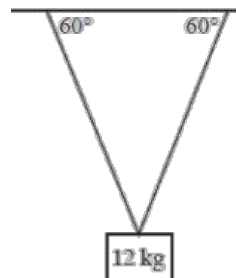


Motion and Forces Topic Test 2

Total Marks: 30

Time: 50 minutes

- 1 A 12kg mass is suspended in equilibrium from a horizontal ceiling by two identical light strings. Each string makes an angle of 60° with the ceiling, as shown. If the force of gravity is $g \text{ m/s}^2$ then what is the magnitude, in newtons, of the tension in each string?



2

- 2 The work done, W , by a constant force, $\vec{F}$, in moving a particle through a displacement, is defined by the formula $W = \vec{F} \cdot \vec{s}$. A force described by the vector $\vec{F} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ moves a particle along the line l from $P(-1, 2)$ to $Q(2, -2)$.

(i). Find $\vec{s} = \overrightarrow{PQ}$ and hence find the value of W .

1

(ii). Hence, verify that W is also given by $W = (\vec{F} \cdot \hat{s})|\vec{s}|$.

1

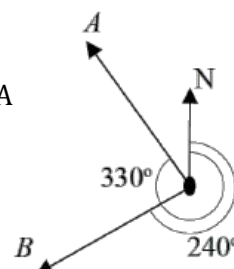
(iii). Find the component of $\vec{F}$ in the direction of l .

1

- 3 The movements of a drone flying straight and level at a constant speed are being monitored via a GPS screen. One unit on the screen represents 100 m in the air. The initial position vector of the drone on the screen is $5\hat{i} + 12\hat{j}$. An hour later its position vector on the screen is $15\hat{i} + 30\hat{j}$. Find the distance travelled by the drone in km, correct to two decimal places.

2

- 4 During a movie scene a camera is attached to three drones by inextensible chords, completely stretched horizontally, with no vertical movement. Drone A is applying a force of 30 N on a bearing of 330° , while drone B is applying a force of 36 N on a bearing of 240° , as shown in the diagram. If the camera is to be kept stationary, calculate the bearing of drone C and the force it needs to apply to the camera, giving your answers to 2 decimal places.



3

- 5 Sienna intends to row her boat from the south bank of a river to meet with her friends on the north bank. The river is 100 metres wide. Sienna's rowing speed is 5 metres per second when the water is still. The river is flowing east at a rate of 4 metres per second. Sienna's boat is also being impacted by a wind blowing from the south-west, which pushed the boat at 8 metres per second. She starts rowing across the river by steering the boat such that it is perpendicular to the south bank.

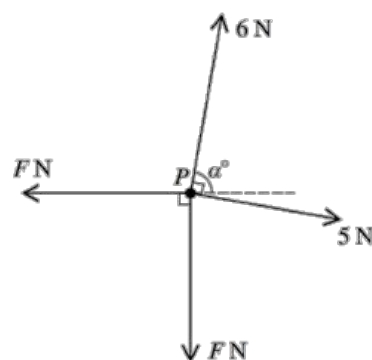
- (i). Show that the velocity of Sienna's boat can be expressed as the component vector: 2

$$(4 + 4\sqrt{2})\underline{i} + (5 + 4\sqrt{2})\underline{j}$$

- (ii). Calculate the speed of the boat, correct to 2 decimal places. 1

- (iii). Determine the distance rowed from Sienna's starting point to her landing point and how long it will take her to reach the north bank. 3

- 6 A particle P is in equilibrium under the action of four forces of magnitude 6 N, 5 N, F N and F N acting in the directions shown in the diagram below.



- (i). By considering the vertical components of the forces, find an expression for F in terms of α . 1

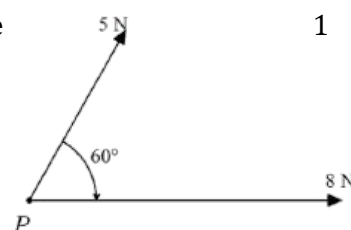
- (ii). Hence, determine the value of α , correct to the nearest degree. 2

- 7 The position of a jet flying straight and level at a constant speed is plotted on its GPS screen. One unit on the screen represents 1 km in the air. At 9:00am the position vector of the plane is $50\underline{i} + 12\underline{j}$. One minute later it is $35\underline{i} + 52\underline{j}$. Find:

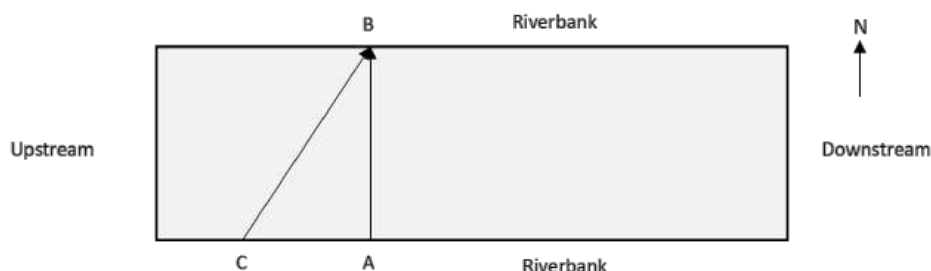
- (i). The position vector at 9:20 am. 2

- (ii). The velocity (as a vector) in km/h. 1

- 8 Forces of magnitude 8 N and 5 N act on a particle P . The angle between the directions of the two forces is 60° as shown in the diagram. What is the magnitude and direction of the resultant force acting on P ?



- 9 Danny wants to swim across the river from position A to position B shown on the diagram below. The riverbanks are parallel straight lines, and $\overline{AB}$ is perpendicular to the riverbanks. The speed of the river flow is 30 m/min from west to east and the river is 240 metres wide. Assume Danny swims at a constant speed of 60 m/min .



- (i). In which direction should Danny aim to swim, in order to move from point A to point B in a straight line under the influence of the current? Draw a vector diagram to demonstrate the velocities and show all necessary working out. 1
- (ii). Gabriel also swims at 60 m/min . but he is to start at position C , which is 60 metres upstream from position A as shown in the diagram. In which direction should Gabriel aim to swim in order to swim directly to position B under the influence of the current? (Draw a vector diagram to demonstrate the velocities and show all necessary working out. Round your answer to one decimal place.) 3

- 10 Fred, Mario and Romeo are fighting over a crown. The three of them are holding on to the crown in the formation as shown in the diagram. If no one manages to pull the crown in their direction (i.e., the crown does not move) and Fred is applying a force of 40N , exactly how much force are Mario and Romeo applying to the crown?

